CHANDERNAGORE COLLEGE

SEMESTER – III B.A. / B.Sc. / B.COM (MAJOR/MINOR/SEC) INTERNAL ASSESSMENT EXAMINATION, 2024

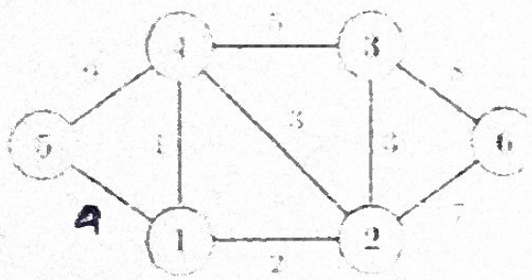
SUBJECT: COMPUTER SCIENCE COURSE CODE: COMP 3011 COURSE NAME: DISCRETE STRUCTURE

FULL MARKS: 10 TIME: 30 MINUTES

ATTEMPT ANY 02 OUT OF THE FOLLOWING 03 QUESTIONS 2x5=10

1) Prove that the maximum number of edges in a simple disconnected graph G with n vertices and k components is (n-k)(n-k+1)/2 edges.

2) Show how Kruskal's algorithm find a minimal spanning tree for the following graph



3) Draw the graph represented by given Adjacency matrix

a)

	A	B	C	D	E
A	0	1	1	0	1
B	1	0	1	0	0
C	1	1	0	1	1
D	0	0	1	0	0
E	1	0	1	0	0

B)

$$\begin{array}{c|ccccc} & A & B & C & E & D \\ \hline A & 0 & 1 & 2 & 2 & 0 \\ B & 1 & 0 & 0 & 0 & 2 \\ C & 2 & 0 & 0 & 1 & 0 \\ D & 2 & 0 & 1 & 0 & 1 \\ E & 0 & 2 & 0 & 1 & 0 \end{array}$$